

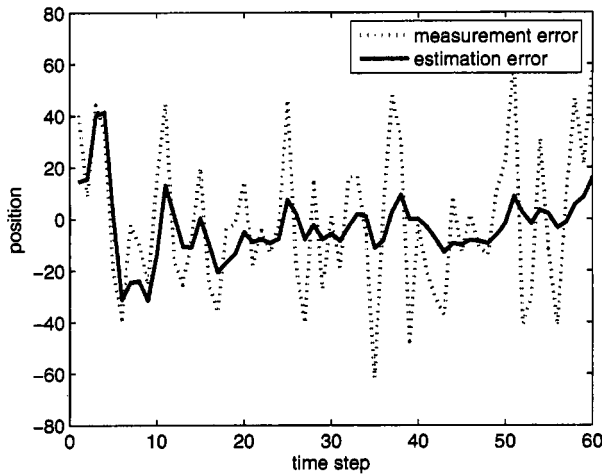


Figure 5.5 The position-measurement error and position estimation error for Example 5.1.

5.4 ALTERNATE PROPAGATION OF COVARIANCE

In this section, we derive an alternate equation for the propagation of the estimation-error covariance P . This alternate equation, based on [Gre01], can be used to find a closed-form equation for a scalar Kalman filter.³ It can also be used to find a fast solution to the steady-state estimation-error covariance.

5.4.1 Multiple state systems

Recall from Equation (5.19) the update equations for the estimation-error covariance:

$$\begin{aligned} P_k^- &= F_{k-1} P_{k-1}^+ F_{k-1}^T + Q_{k-1} \\ P_k^+ &= P_k^- - P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} H_k P_k^- \end{aligned} \quad (5.39)$$

If the $n \times n$ matrix P_k^- can be factored as

$$P_k^- = A_k B_k^{-1} \quad (5.40)$$

where A_k and B_k are $n \times n$ matrices to be determined, then P_{k+1}^- satisfies

$$P_{k+1}^- = A_{k+1} B_{k+1}^{-1} \quad (5.41)$$

where A and B are propagated as follows:

$$\begin{bmatrix} A_{k+1} \\ B_{k+1} \end{bmatrix} = \begin{bmatrix} (F_k + Q_k F_k^{-T} H_k^T R_k^{-1} H_k) & Q_k F_k^{-T} \\ F_k^{-T} H_k^T R_k^{-1} H_k & F_k^{-T} \end{bmatrix} \begin{bmatrix} A_k \\ B_k \end{bmatrix} \quad (5.42)$$

³The equations given in [Gre01] have some typographical errors that have been corrected in this section.

This can be seen by noting from Equation (5.42) that

$$\begin{aligned} B_{k+1}^{-1} &= [F_k^{-T} H_k^T R_k^{-1} H_k A_k + F_k^{-T} B_k]^{-1} \\ &= [F_k^{-T} (H_k^T R_k^{-1} H_k A_k B_k^{-1} + I) B_k]^{-1} \\ &= B_k^{-1} [H_k^T R_k^{-1} H_k A_k B_k^{-1} + I]^{-1} F_k^T \end{aligned} \quad (5.43)$$

From Equation (5.42) we see that

$$A_{k+1} B_{k+1}^{-1} = [(F_k + Q_k F_k^{-T} H_k^T R_k^{-1} H_k) A_k + Q_k F_k^{-T} B_k] B_{k+1}^{-1} \quad (5.44)$$

Substituting the expression for B_{k+1}^{-1} into this equation gives

$$\begin{aligned} A_{k+1} B_{k+1}^{-1} &= [(F_k + Q_k F_k^{-T} H_k^T R_k^{-1} H_k) A_k + Q_k F_k^{-T} B_k] \times \\ &\quad B_k^{-1} [H_k^T R_k^{-1} H_k A_k B_k^{-1} + I]^{-1} F_k^T \\ &= [(F_k + Q_k F_k^{-T} H_k^T R_k^{-1} H_k) A_k B_k^{-1} + Q_k F_k^{-T}] \times \\ &\quad [H_k^T R_k^{-1} H_k A_k B_k^{-1} + I]^{-1} F_k^T \end{aligned} \quad (5.45)$$

Substituting P_k^- for $A_k B_k^{-1}$ in the above equation gives

$$\begin{aligned} A_{k+1} B_{k+1}^{-1} &= [(F_k + Q_k F_k^{-T} H_k^T R_k^{-1} H_k) P_k^- + Q_k F_k^{-T}] \times \\ &\quad [H_k^T R_k^{-1} H_k P_k^- + I]^{-1} F_k^T \\ &= [F_k P_k^- + Q_k F_k^{-T} (H_k^T R_k^{-1} H_k P_k^- + I)] \times \\ &\quad [H_k^T R_k^{-1} H_k P_k^- + I]^{-1} F_k^T \\ &= F_k P_k^- [H_k^T R_k^{-1} H_k P_k^- + I]^{-1} F_k^T + Q_k F_k^{-T} F_k^T \end{aligned} \quad (5.46)$$

Applying the matrix inversion lemma to the term in brackets gives

$$\begin{aligned} A_{k+1} B_{k+1}^{-1} &= F_k P_k^- [I - H_k^T (H_k P_k^- H_k^T + R_k)^{-1} H_k P_k^-] F_k^T + Q_k \\ &= F_k [P_k^- - P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} H_k P_k^-] F_k^T + Q_k \\ &= F_k P_k^+ F_k^T + Q_k \\ &= P_{k+1}^- \end{aligned} \quad (5.47)$$

So we see that $A_{k+1} B_{k+1}^{-1} = P_{k+1}^-$.

Equation (5.42) can be used to obtain a quick solution to the steady-state covariance for multidimensional systems (although not a closed-form solution). Suppose that F , Q , H , and R are constant matrices. From Equation (5.42) we obtain

$$\begin{aligned} \begin{bmatrix} A_{k+1} \\ B_{k+1} \end{bmatrix} &= \begin{bmatrix} (F + Q F^{-T} H^T R^{-1} H) & Q F^{-T} \\ F^{-T} H^T R^{-1} H & F^{-T} \end{bmatrix} \begin{bmatrix} A_k \\ B_k \end{bmatrix} \\ &= \Psi \begin{bmatrix} A_k \\ B_k \end{bmatrix} \\ \begin{bmatrix} A_k \\ B_k \end{bmatrix} &= \Psi^{k-1} \begin{bmatrix} P_1^- \\ I \end{bmatrix} \end{aligned} \quad (5.48)$$

where we used the fact that $A_1 = P_1^-$ and $B_1 = I$ satisfies the original factoring of Equation (5.40). Now we can successively square Ψ a total of p times to obtain Ψ^2 , Ψ^4 , Ψ^8 , and so on, until Ψ^{2^p} converges to a steady-state value:

$$\begin{bmatrix} A_\infty \\ B_\infty \end{bmatrix} \approx \Psi^{2^p} \begin{bmatrix} P_1^- \\ I \end{bmatrix} \quad \text{for large } p \quad (5.49)$$

The steady-state covariance is $P_\infty^- = A_\infty B_\infty^{-1}$. We can also find the steady-state Kalman gain by simply iterating the filter equations from Equation (5.19), but the method in this section could be a much quicker way to find the steady-state gain. Once we find P_∞^- as shown above, we compute $K_\infty = P_\infty^- H^T (H P_\infty^- H^T + R)^{-1}$ as the steady-state Kalman filter gain. More discussion of steady-state Kalman filtering is given in Section 7.3.

5.4.2 Scalar systems

Equation (5.42) can be used to obtain a closed-form solution for the scalar Kalman filter for time-invariant systems. Suppose that F , Q , H , and R are constant scalars. Then from Equation (5.42) we obtain

$$\begin{aligned} \begin{bmatrix} A_{k+1} \\ B_{k+1} \end{bmatrix} &= \begin{bmatrix} F + \frac{H^2 Q}{FR} & \frac{Q}{F} \\ \frac{H^2}{FR} & \frac{1}{F} \end{bmatrix} \begin{bmatrix} A_k \\ B_k \end{bmatrix} \\ &= \Psi \begin{bmatrix} A_k \\ B_k \end{bmatrix} \end{aligned} \quad (5.50)$$

where Ψ is defined by the above equation. Now find the eigendata of Ψ . Suppose that the eigenvalues of Ψ are λ_1 and λ_2 , and the eigenvectors of Ψ are combined to create the 2×2 matrix M . Then

$$\Psi = M \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} M^{-1} \quad (5.51)$$

and we obtain

$$\begin{aligned} \begin{bmatrix} A_k \\ B_k \end{bmatrix} &= \Psi^{k-1} \begin{bmatrix} A_1 \\ B_1 \end{bmatrix} \\ &= M \begin{bmatrix} \lambda_1^{k-1} & 0 \\ 0 & \lambda_2^{k-1} \end{bmatrix} M^{-1} \begin{bmatrix} P_1^- \\ 1 \end{bmatrix} \end{aligned} \quad (5.52)$$

where we used the fact that $A_1 = P_1^-$ and $B_1 = 1$ satisfies the original factoring of Equation (5.40). Working through the math to obtain λ_1 , λ_2 , and M gives the following.

$$\begin{aligned} P_k^- &= \frac{\tau_1 \mu_1^{k-1} (2RH^2 P_1^- - \tau_2) - \tau_2 \mu_2^{k-1} (2H^2 P_1^- - \tau_1)}{2H^2 \mu_1^{k-1} (2RH^2 P_1^- - \tau_2) - 2H^2 \mu_2^{k-1} (2H^2 P_1^- - \tau_1)} \\ \lambda_1 &= \frac{H^2 Q + R(F^2 + 1) + \sigma}{2FR} \\ \lambda_2 &= \frac{H^2 Q + R(F^2 + 1) - \sigma}{2FR} \\ \sigma &= \sqrt{H^2 Q + R(F+1)^2} \sqrt{H^2 Q + R(F-1)^2} \\ \tau_1 &= H^2 Q + R(F^2 - 1) + \sigma \\ \tau_2 &= H^2 Q + R(F^2 - 1) - \sigma \\ \mu_1 &= H^2 Q + R(F^2 + 1) + \sigma \\ \mu_2 &= H^2 Q + R(F^2 + 1) - \sigma \end{aligned}$$

$$\begin{aligned}
M &= \begin{bmatrix} \frac{\tau_1}{2H^2} & \frac{\tau_2}{2H^2} \\ 1 & 1 \end{bmatrix} \\
M^{-1} &= \frac{1}{\tau_1(R-1) + 2\sigma} \begin{bmatrix} 2RH^2 & -\tau_1 \\ -2RH^2 & R\tau_1 \end{bmatrix}
\end{aligned} \tag{5.53}$$

This is a closed-form equation for the time-varying Kalman filter for a scalar time-invariant system. This can easily be used to obtain the steady-state value of P_k^- . Note that $\mu_2 < \mu_1$. As k increases, μ_2^k gets smaller and smaller relative to μ_1^k . Therefore

$$\begin{aligned}
\lim_{k \rightarrow \infty} P_k^- &= \lim_{k \rightarrow \infty} \frac{\tau_1 \mu_1^{k-1} (2RH^2 P_1^- - \tau_2) - \tau_2 \mu_2^{k-1} (2H^2 P_1^- - \tau_1)}{2H^2 \mu_1^{k-1} (2RH^2 P_1^- - \tau_2) - 2H^2 \mu_2^{k-1} (2H^2 P_1^- - \tau_1)} \\
&= \lim_{k \rightarrow \infty} \frac{\tau_1 \mu_1^{k-1} (2RH^2 P_1^- - \tau_2)}{2H^2 \mu_1^{k-1} (2RH^2 P_1^- - \tau_2)} \\
&= \frac{\tau_1}{2H^2}
\end{aligned} \tag{5.54}$$

This gives the steady-state covariance for a scalar system.

■ EXAMPLE 5.2

In this example, we will show how a scalar covariance can be propagated. Consider the following scalar system:

$$\begin{aligned}
x_{k+1} &= x_k + w_k \\
y_k &= x_k + v_k \\
w_k &\sim (0, 1) \\
v_k &\sim (0, 1)
\end{aligned} \tag{5.55}$$

This is a very simple system but one that arises in many applications. For example, it may represent some slowly varying parameter x_k that we measure directly. The process noise term w_k accounts for the variations in x_k , and the measurement noise term v_k accounts for measurement errors. In this system, we have $F = H = Q = R = 1$. Substituting these values in Equation (5.53) gives

$$\begin{aligned}
\tau_1 &= 1 + \sqrt{5} \\
\tau_2 &= 1 - \sqrt{5} \\
\mu_1 &= 3 + \sqrt{5} \\
\mu_2 &= 3 - \sqrt{5} \\
P_k^- &= \frac{\tau_1 \mu_1^{k-1} (2P_1^- - \tau_2) - \tau_2 \mu_2^{k-1} (2P_1^- - \tau_1)}{2\mu_1^{k-1} (2P_1^- - \tau_2) - 2\mu_2^{k-1} (2P_1^- - \tau_1)}
\end{aligned} \tag{5.56}$$

Taking the limit as $k \rightarrow \infty$ gives the steady-state value of P_k^- :

$$\begin{aligned}
P_\infty^- &= \frac{\tau_1}{2} \\
&= \frac{1 + \sqrt{5}}{2} \\
&\approx 1.62
\end{aligned} \tag{5.57}$$